

● MEDIUM

● ALGEBRA

# Linear equations in two variables

01

10 questions · ~15 min

Q1

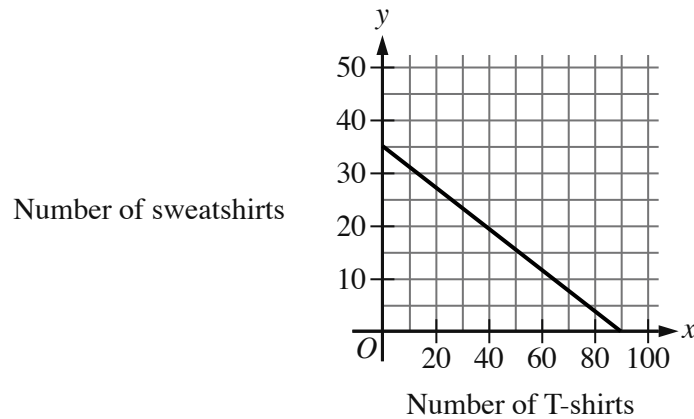
GRID-IN ALGEBRA

Line  $k$  is defined by  $y = -\frac{17}{3}x + 5$ . Line  $j$  is perpendicular to line  $k$  in the  $xy$ -plane.  
What is the slope of line  $j$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (0 comma 35)
  - (90 comma 0)

The graph models the relationship between the number of T-shirts,  $x$ , and the number of sweatshirts,  $y$ , that Kira can purchase for a school fundraiser. Which equation could represent this relationship?

- A.  $y = 7x + 18$
- B.  $7x + 18y = 630$
- C.  $y = 18x + 7$
- D.  $18x + 7y = 630$

Line  $p$  is defined by  $2y + 18x = 9$ . Line  $r$  is perpendicular to line  $p$  in the  $xy$ -plane. What is the slope of line  $r$ ?

A.  $-9$

B.  $-\frac{1}{9}$

C.  $\frac{1}{9}$

D.  $9$

A neighborhood consists of a 2-hectare park and a 35-hectare residential area. The total number of trees in the neighborhood is 3,934. The equation  $2x + 35y = 3,934$  represents this situation. Which of the following is the best interpretation of  $x$  in this context?

- A. The average number of trees per hectare in the park
- B. The average number of trees per hectare in the residential area
- C. The total number of trees in the park
- D. The total number of trees in the residential area

Line  $k$  is defined by  $y = \frac{17}{7}x + 4$ . Line  $j$  is parallel to line  $k$  in the  $xy$ -plane. What is the slope of line  $j$ ?

A.  $\frac{7}{17}$

B.  $\frac{17}{7}$

C. 4

D. 17

A batch of banana milkshakes consists of 4 cups of ice cream and 2 bananas and has 1,114 milligrams mg of calcium. There is 276 mg of calcium in 1 cup of the ice cream used to make this batch of milkshakes. How much calcium, in mg, is in 1 banana?

- A. 5
- B. 10
- C. 419
- D. 1,104

Q7

GRID-IN ALGEBRA

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of \$ 641, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil,  $x$ , and the number of tons of crushed stone,  $y$ , Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



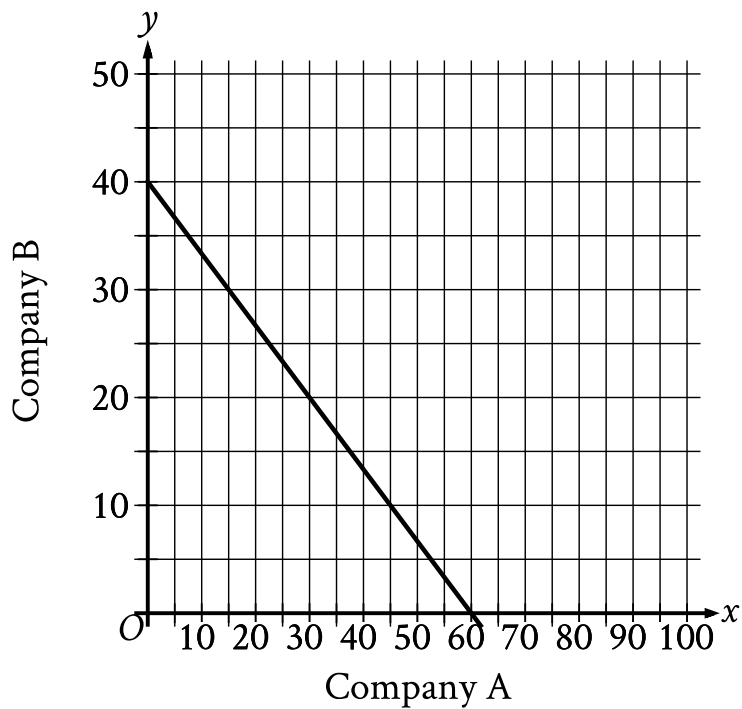
When line  $n$  is graphed in the  $xy$ -plane, it has an  $x$ -intercept of  $-4, 0$  and a  $y$ -intercept of  $0, \frac{86}{3}$ . What is the slope of line  $n$ ?

A.  $\frac{3}{344}$

B.  $\frac{6}{43}$

C.  $\frac{43}{6}$

D.  $\frac{344}{3}$



- The line slants down from left to right.
- The line passes through the following points:
  - (0 comma 40)
  - (60 comma 0)

The graph shows the relationship between the number of shares of stock from Company A,  $x$ , and the number of shares of stock from Company B,  $y$ , that Simone can purchase. Which equation could represent this relationship?

- A.  $y = 8x + 12$
- B.  $8x + 12y = 480$
- C.  $y = 12x + 8$
- D.  $12x + 8y = 480$

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is 63 inches. The equation  $3x + 6y = 63$  represents this situation, where  $x$  is the number of sides of figure A and  $y$  is the number of sides of figure B. Which statement is the best interpretation of 6 in this context?

- A. Each side of figure B has a length of 6 inches.
- B. The number of sides of figure B is 6.
- C. Each side of figure A has a length of 6 inches.
- D. The number of sides of figure A is 6.